Abenezer Bekele

APPLIED DATA SCIENTIST

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PROFESSIONAL SUMMARY

Detail oriented Computer Science graduate with hands-on experience building ETL/ELT pipelines, deploying machine learning models (Random Forest, Decision Trees, PCA, Logistic Regression), and developing data integration solutions. Proficient in Python (Pandas, NumPy, Scikit-learn) and SQL for data transformation, statistical analysis, and workflow automation. Demonstrated foundation in linear algebra, probability, and optimization through applied ML projects. Experienced with AWS, Docker, CI/CD, and Agile/Scrum. Proven ability to communicate data driven insights to non technical stakeholders. U.S. Naturalized Citizen , no sponsorship required.

EDUCATION

University of Wisconsin – River Falls | River Falls, WI

- **Bachelor of Science in Computer Science and Information Systems | December 2024**
  Relevant Coursework: Database Systems, Data Science, Software Engineering, Object-Oriented Programming, Data Structure Algorithms

CERTIFICATIONS

- IBM data science professional certificate
- Docker Foundations Professional Certificate
- AWS Cloud Practitioner Essentials (In Progress)

WORK EXPERIENCE

Development Intern | A Better Road Foundation **2025**

- Designed, developed, and deployed end-to-end web platform; analyzed datasets using Google Analytics to surface KPI insights for leadership decision-making — directly informing content and email engagement strategies.
- Acted as sole technical liaison between non-technical stakeholders and the IS team; gathered requirements, translated organizational goals into functional solutions, and documented all data interfaces for auditability.
- Built custom interactive elements using JavaScript (Velo), ensuring data integrity and responsive user experience across platforms.
- Implemented SEO best practices and established repeatable, scalable web data reporting workflows for the organization

TECHNICAL SKILLS

- **Applied ML & Data Science:** Machine Learning (Scikit-learn, Random Forest, Decision Trees, Logistic Regression), Predictive Modeling, Hyperparameter Tuning, Statistical Analysis, Exploratory Data Analysis (EDA)
- **NLP / LLM Awareness:** Generative AI integration (Google Gemini Pro), Metadata extraction from unstructured data, AI-powered workflow automation
- **Languages & Programming:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL (PostgreSQL, MS SQL Server, MySQL), R, JavaScript, TypeScript, Bash Scripting
- **Data & ML Ops:** Google BigQuery , AWS (EC2, S3, Lambda, API Gateway), Docker, CI/CD Pipelines, Azure DevOps, Git/GitHub, Jupyter Notebooks, Unit Testing (PyTest)
- **Databases & Data Warehousing:** PostgreSQL, MS SQL Server, MySQL, MongoDB, DynamoDB, Snowflake (familiar), CTEs, Window Functions, Stored Procedures, Query Optimization, Dimensional Modeling (Star/Snowflake)
- **BI & Visualization:** Power BI (DAX), Tableau, Matplotlib, Seaborn, Excel (Pivot Tables, Power Query, VBA/Macros)
- **Agile & Process:** Agile/Scrum, Jira, Code Reviews, Technical Documentation, Stakeholder Collaboration, Cross-functional Communication

PROJECTS

PCA & Dimensionality Reduction Analysis | GitHub

Python, Scikit-learn, NumPy, Matplotlib, Seaborn, Jupyter

- Applied Principal Component Analysis (PCA) for dimensionality reduction on a multi-feature dataset; identified optimal component count by analyzing explained variance ratios demonstrating applied linear algebra and statistical modeling foundations.
- Visualized component loadings and cumulative variance curves using Matplotlib/Seaborn to communicate feature importance and data structure to non-technical stakeholders.
- Reduced high-dimensional feature space while preserving dataset variance, improving downstream model efficiency a core technique in search ranking and recommendation system pipelines.

Drug Recommendation: Decision Tree Classifier | GitHub

Python, Scikit-learn, Pandas, Matplotlib, Jupyter

- Trained and evaluated a Decision Tree classifier to rank and score drug recommendations based on patient attributes (age, BP, cholesterol, Na/K ratio); applied label encoding, feature scaling, and hyperparameter tuning.
- Visualized decision tree structure and feature importance scores to produce interpretable, explainable AI insights — a critical requirement in compliance-sensitive and high-stakes decision domains.
- Evaluated model performance using accuracy scores and confusion matrices; iterated on depth/split parameters to optimize predictive accuracy without overfitting.

Automated ETL & ML Pipeline: Titanic Survival Prediction | GitHub

Python, Scikit-learn, Pandas, PyTest, Jupyter

- Built a robust ETL pipeline handling missing values, feature scaling, and normalization to produce model-ready datasets — applying best practices in data wrangling and quality assurance.
- Developed and evaluated end-to-end ML models (Random Forest, Decision Trees, Logistic Regression) using Scikit-learn; optimized predictive accuracy through hyperparameter tuning.
- Applied unit testing (PyTest), source control (Git), and modular coding practices to ensure pipeline reusability and maintainability — aligned with CI/CD best practices.

Relational Database Design & SQL Optimization (COVID-19) | GitHub

SQL, MS SQL Server, PostgreSQL

- Authored 12+ complex SQL queries using Window Functions, CTEs, and Self-Joins to transform raw global health data into structured analytical views for stakeholder reporting.
- Optimized query execution through strategic indexing and stored procedures; applied NULL filtering and data validation constraints to enforce data integrity standards.

Public Interest Data Pipeline: Chicago Urban Analysis | GitHub

Python, Pandas, Jupyter Notebooks

- Engineered a data pipeline to process diverse urban datasets (socio-economic, education, population) to identify service gaps in metropolitan areas.
- Cleaned and normalized 'messy' public data; developed automated reporting modules to communicate demographic trends clearly to non-technical stakeholders.

FinTracker: AI-Powered Financial API | GitHub

Node.js, PostgreSQL, Google Gemini Pro, Docker

- Deployed a production RESTful API with JWT authentication and role-based access control handling live financial transaction data; containerized using Docker for consistent, scalable deployment.
- Leveraged Generative AI (Google Gemini Pro Vision API) to automate metadata extraction from unstructured images, building an ETL-style data ingestion pipeline.